



MSC MECHANICAL ENGINEERING

Imaging in Fluid Dynamics

Experiment 4 – Fluid-Structure Interaction

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1 Abstract

This experiment investigates the fluid–structure interaction of flexible sheets exposed to a uniform airflow in a wind tunnel. The sheets, clamped at their base and free at their tip, undergo static reconfiguration and, above a critical velocity, fluttering oscillations. High-speed imaging and image-based tracking are used to characterize both the deformation and the dynamic response of the sheets. Experiments were conducted on sheets of varying thicknesses ($h = 100, 300, \text{ and } 500 \mu\text{m}$), with two sheets tested for each thickness: one short and one long. The airflow was controlled by varying the wind tunnel motor frequency from 10 to 40 Hz, resulting in freestream velocities ranging from $U \simeq 7.2$ to 29.7 ms^{-1} and corresponding Cauchy numbers from $Ca \simeq 0.55$ to 112. The static deformation was quantified using the reconfiguration number, which decreases from $R \simeq 0.92$ to 0.34 with increasing Cauchy number.

The fluttering dynamics were analyzed from the sheet tip motion using Fourier spectral analysis and temporal autocorrelation. For long sheets, the larger oscillation amplitudes lead to a clear spectral peak and a well-defined autocorrelation, allowing accurate extraction of the dominant flutter frequency. In contrast, short sheets exhibit smaller-amplitude oscillations, and above a wind tunnel frequency of 10 Hz the signal becomes increasingly noisy, preventing reliable frequency extraction from the Fourier spectrum. At sufficiently low velocities, however, both methods yield consistent frequencies in agreement with theoretical expectations.

Overall, the results confirm that the Cauchy number governs both the static reconfiguration and the dynamic fluttering of flexible sheets, while emphasizing the role of sheet length and oscillation amplitude in the accuracy of frequency identification.

2 Introduction

Fluid–structure interaction (FSI) is a fundamental phenomenon encountered in a wide range of natural and engineering systems, where fluid forces interact with deformable structures and may lead to large deformations or dynamic instabilities. In natural environments, the interaction between wind and vegetation plays a key role in determining aerodynamic loads, bending, and drag reduction through reconfiguration. These mechanisms have been extensively studied in the context of plant–flow interaction [2].

In engineering applications, fluid–structure interaction is a major concern for slender structures such as bridges, towers, and high-rise buildings exposed to wind. A well-known illustration of flow-induced instability is the collapse of the *Tacoma Narrows Bridge* in 1940, which highlighted the importance of accounting for aerodynamic coupling and resonance effects in structural design.

Fluid–structure interaction is also highly relevant in sports applications, particularly in sailing, where the deformation of sails under wind loading directly affects aerodynamic performance and stability. The dynamics of flexible plates and membranes in a flow, including flutter and flapping instabilities, have been widely investigated in the literature [3].

Motivated by these examples, the present experiment focuses on the study of fluid–structure interaction in a controlled and simplified configuration, allowing the fundamental physical mechanisms to be isolated and analyzed quantitatively. The experimental setup and measurement methodology are described in the following sections.



(Taccoma Bridge collapse 1940)



(America's cup 2013)

Figure 1: Example of FSI in real life

3 Theoretical Background

3.1 Bending of a Slender Beam

The deformation of a flexible sheet subjected to a fluid flow can be modeled, in a first approximation, as the bending of a slender beam. Under the assumption of small thickness compared to its length, the deformation is dominated by bending moments and can be described using Euler–Bernoulli beam theory. The elastic response of the sheet is characterized by the bending moment M , the shear force V , and the resultant transverse force F , which are related to the local curvature $\kappa(s)$ along the sheet by

$$\begin{aligned} M(s) &= EI \kappa(s), \\ V(s) &= \frac{dM}{ds} = EI \frac{d\kappa}{ds}, \\ F &= V(0). \end{aligned} \quad (1)$$

Here, E denotes the Young's modulus of the material and I is the second moment of area of the sheet cross-section, which for a rectangular sheet of width b and thickness h is given by

$$I = \frac{bh^3}{12}. \quad (2)$$

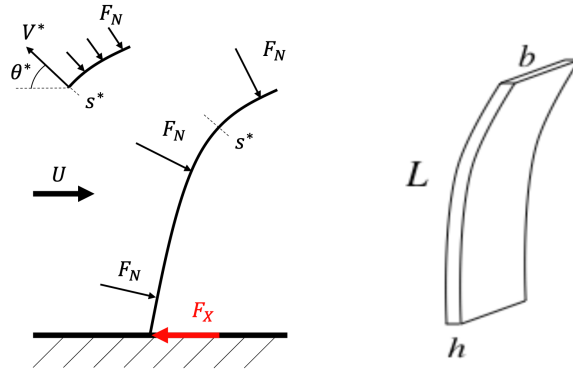


Figure 2: Free-body diagram of the beam.

The bending moment M quantifies the elastic resistance to curvature, the shear force V represents the internal transverse force within the sheet, and the force F corresponds to the total transverse load exerted by the fluid at the clamped base. This simplified beam model provides a direct link between the elastic properties of the sheet and its deformation under aerodynamic loading.

3.2 Cauchy Number

The relative importance of fluid forces compared to elastic restoring forces is quantified by the Cauchy number. It is defined as the ratio between inertial forces induced by the flow and elastic forces resisting deformation:

$$Ca = \frac{\rho U^2 b L^3}{EI}, \quad (3)$$

where ρ is the fluid density, U is the freestream velocity, and L is the sheet length. A low Cauchy number corresponds to a rigid behavior dominated by elasticity, while a high Cauchy number indicates a highly flexible structure strongly deformed by the flow.

3.3 Reconfiguration Number

As the sheet bends under the action of the flow, its effective frontal area decreases, leading to a reduction in aerodynamic drag. This effect is characterized by the reconfiguration number, defined as the ratio between the drag force acting on the flexible sheet and the drag force that would act on an equivalent rigid sheet:

$$R = \frac{F_D}{\frac{1}{2} C_{D,0} \rho U^2 b L}, \quad (4)$$

where $C_{D,0}$ is the drag coefficient of the rigid sheet. In practice, the reconfiguration number can be approximated by the ratio between the bent length and the undeformed length of the sheet, $R \simeq L_{\text{bent}}/L$. As the Cauchy number increases, the reconfiguration number decreases, reflecting enhanced streamlining and drag reduction.

3.4 Temporal Autocorrelation and Fourier Spectral Analysis

The oscillatory motion of the sheet tip is analyzed using both temporal autocorrelation and Fourier spectral analysis, which provide complementary approaches for frequency extraction. For a discrete time signal $x(t)$, the normalized temporal autocorrelation function of a discrete signal a_j is defined as

$$\rho(t) = \frac{1}{N-t} \frac{\sum_{j=0}^{N-t} (a_j - \bar{a})(a_{j+t} - \bar{a})}{\text{Var}(a)}, \quad (5)$$

where $\{a_j\} = a_0, \dots, a_N$ denotes the discrete signal sequence, $\bar{a}$ is the signal mean, and $\text{Var}(a)$ is the signal variance. Periodic or quasi-periodic oscillations result in repeating peaks in the autocorrelation function, and the dominant oscillation frequency can be obtained from the time lag of the first peak away from zero.

Fourier spectral analysis provides an alternative frequency-domain representation of the signal. By computing the discrete Fourier transform, the time signal is decomposed into its harmonic components, and the dominant oscillation frequency appears as a peak in the frequency spectrum. Fourier analysis is particularly effective for stationary signals with well-defined periodicity, but its accuracy decreases for non-stationary or strongly intermittent dynamics.

3.5 Strouhal Number

The fluttering oscillations of the sheet can be characterized in a dimensionless form using the Strouhal number, defined as

$$St = \frac{fL}{U}, \quad (6)$$

where f is the dominant oscillation frequency. The Strouhal number allows comparison of oscillatory behavior across different flow velocities and sheet configurations and provides insight into the transition between elasticity-dominated and flow-dominated regimes.

4 Experimental Setup

The experiment was conducted in a wind tunnel in which the fluid–structure interaction of a single flexible sheet subjected to an airflow was investigated. The velocity of the wind tunnel was controlled using the frequency control unit of the motor, with the rotation frequency ranging from 0 to 40 Hz; however, the motor rotation becomes unsteady below 10 Hz. To achieve lower airflow velocities, a throttle built into the exhaust of the wind tunnel was used. The velocity characteristic of the wind tunnel was assumed to be calibrated as shown in the Fig. 5, allowing the desired flow velocity to be set by adjusting the rotor frequency proportionally. The flexible sheets were mounted vertically in the test section and clamped at their base using a 3D-printed mount. Fluttering oscillations were investigated using both sheets of 300 μm and 100 μm , avoiding conditions where the sheet would hit the tunnel wall. The deformation and motion of the sheets were recorded using a Phantom Miro high-speed camera operated with the PCC software. A high-intensity flashlight was used to improve the visibility of the white markers on the sheet, allowing for more reliable point tracking.

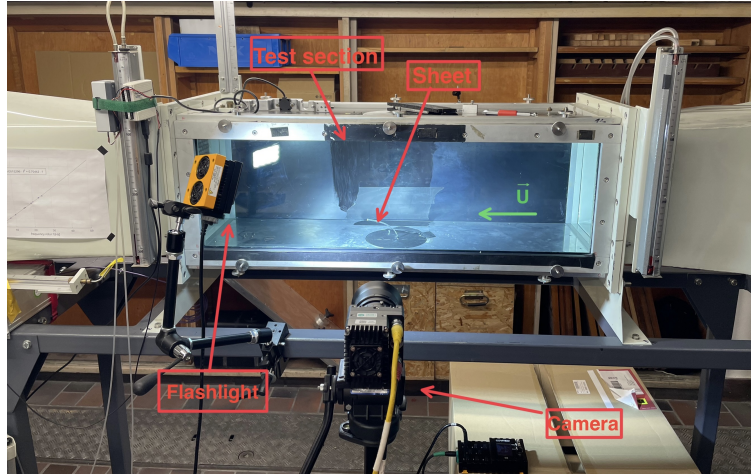


Figure 3: Experimental setup photo

5 Methods and Data Processing

5.1 Experimental Procedure

The flexible sheets used in this experiment were manually cut to the desired dimensions prior to the measurements. For sheet thicknesses of 300 μm and 500 μm , the samples were prepared with a fixed aspect ratio of 10. For the thinnest sheets ($h = 100 \mu\text{m}$), no aspect ratio constraint was imposed.

Instead a constant width of 1.3 cm was used for all samples of this thickness. Once prepared, white markers were placed along the side of each sheet to facilitate point tracking during image processing. Each sheet was mounted vertically on a plastic support and centered in the wind tunnel test section. After ensuring proper alignment, the wind tunnel was switched on and set to the desired motor frequency. The camera parameters were then adjusted, including the exposure time, the acquisition frame rate, and the total duration of the recording. Once these parameters were fixed, image sequences of the sheet deformation and motion were recorded.

Table 1: Summary of experimental configurations and acquisition parameters.

Thickness (μm)	Length (cm)	Width (cm)	AR	Motor frequency (Hz)	FPS / Exposure
100	3.5	1.3	–	10, 20	300 / 300
100	6.0	1.3	–	10, 20	300 / 300
100	10.0	1.3	–	10	300 / 300
300	6.0	0.60	10	10, 20, 30	300 / 300
300	10.0	1.00	10	10, 20, 30	300 / 300
500	6.0	0.60	10	10, 20, 30, 40	300 / 300
500	10.0	1.00	10	10, 20, 30	300 / 300

5.2 Calibration and Data Processing

The spatial calibration was performed using a standard chessboard calibration target placed in the field of view of the camera. This procedure allowed the conversion of pixel coordinates into physical units. From the calibration, a constant conversion factor of

$$1 \text{ px} = 1.770732 \times 10^{-4} \text{ m} \quad (7)$$

was obtained and used consistently for all datasets. An example of the calibration and tracking procedure is shown in Fig. 4. The material properties used for the data analysis were assumed to be constant for all samples. The Young’s modulus of the flexible sheets was taken as 2.3 GPa, and the air density was assumed to be 1.2 kg m^{-3} . These values were used for the computation of the Cauchy number, the natural frequency, and other derived quantities.

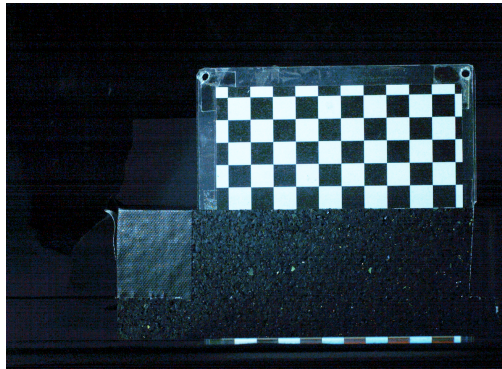


Figure 4: Picture of the calibration chessboard

The video analysis was carried out using MATLAB. A custom tracking code was developed to detect and follow the white markers placed along the side of the sheets, allowing the time-resolved displacement of the sheet tip to be extracted. In addition to tip tracking, the detected marker

positions were used to reconstruct the instantaneous shape of the sheet. At each time step, the sheet profile was fitted using a smooth interpolating polynomial based on Lagrange interpolation, enabling the bending of the sheet to be analyzed in real time.

From the tip position, the effective bent length L_{bent} was extracted and used to compute the reconfiguration number R . The time series obtained from the tracked tip motion were further analyzed using temporal autocorrelation and Fourier spectral analysis to identify the dominant flutter frequencies.

Uncertainty and Sources of Error

Several sources of uncertainty affect the experimental measurements and derived quantities. First, the sheet length was measured with an accuracy of approximately ± 1 mm, which directly impacts the estimation of dimensionless numbers involving the characteristic length. The sheet was assumed to be centered in the wind tunnel test section; however, in practice, the positioning was performed manually. Due to variations in the effective clamping width and the manual alignment procedure, the centering accuracy is estimated to be on the order of ± 0.5 cm. This positional uncertainty leads to an additional error in the estimation of the local flow velocity, as different calibration relationships are used depending on whether the sheet is located at the center or displaced laterally.

In addition, velocity fluctuations in the wind tunnel contribute to uncertainty in the determination of the freestream velocity. The root-mean-square of the velocity fluctuations, u'_{rms} , was evaluated and compared to the mean flow velocity U . As shown in Fig. 5, the turbulence intensity u'_{rms}/U decreases significantly with increasing flow velocity, reaching values close to 1% at the highest speeds. While these fluctuations are relatively small at high velocities, they may affect frequency extraction and reconfiguration measurements at lower flow speeds, where turbulence intensity is larger.

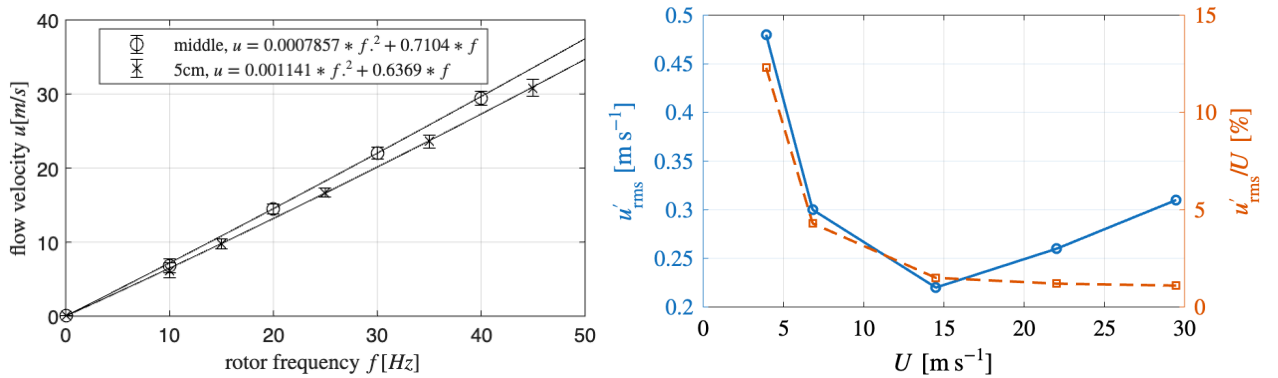


Figure 5: Velocity fluctuation measurements in the wind tunnel.

6 Results

Table 2: Summary of measured results for each run. The sheet thickness t is given in μm , the length L and width b in cm, the motor frequency f_M , the extracted frequencies f_{FFT} and f_{AC} in Hz, the freestream velocity U in m/s, and the dimensionless Cauchy number Ca , reconfiguration number R , and Strouhal numbers St_{FFT} and St_{AC} .

t	L	b	AR	f_M	U	Ca	R	f_{FFT}	f_{AC}	St_{FFT}	St_{AC}
300	6.0	0.60	10	10	7.18	2.58	0.877	1.00*	20.0	0.008	0.167
300	6.0	0.60	10	20	14.5	10.6	0.676	22.97	27.27	0.095	0.113
300	6.0	0.60	10	30	22.0	24.3	0.536	1.12*	37.5	0.003	0.102
300	10.0	1.00	10	10	7.18	12.0	0.755	9.97	9.68	0.139	0.135
300	10.0	1.00	10	20	14.5	48.9	0.495	0.67*	18.75	0.005	0.130
300	10.0	1.00	10	30	22.0	112	0.346	19.31*	33.33	0.088	0.151
500	6.0	0.60	10	10	7.18	0.56	0.926	58.80*	75.00*	0.491	0.627
500	6.0	0.60	10	20	14.5	2.30	0.822	0.33*	37.5	0.001	0.155
500	6.0	0.60	10	30	22.0	5.28	0.689	40.95	42.86	0.112	0.117
500	6.0	0.60	10	40	29.5	9.42	0.575	1.00*	42.86	0.002	0.087
500	10.0	1.00	10	10	7.18	2.61	0.848	15.95	13.64	0.222	0.190
500	10.0	1.00	10	20	14.5	10.6	0.673	2.66*	18.75	0.018	0.129
500	10.0	1.00	10	30	22.0	24.3	0.521	0.33*	21.43	0.002	0.097

*Frequency values dominated by measurement noise due to negligible oscillatory motion; values are not representative of a physical flutter mode.

6.1 Reconfiguration Number vs Cauchy Number

Data extraction

The Cauchy number was computed from the motor frequency applied to the wind tunnel, which was converted into a freestream velocity using the wind tunnel calibration. In all cases, the sheet was positioned at the center of the test section to ensure consistent flow conditions. The reconfiguration number was obtained from the effective bent length L_{bent} , extracted from the reconstructed sheet shape. For each run, L_{bent} was averaged over the entire oscillation time series in order to estimate a mean reconfiguration number representative of the configuration. By plotting the values obtained for all samples, the relationship between the reconfiguration number and the Cauchy number was established, as shown in Fig. 6. A linear fit in logarithmic scale was applied to the data in the variable regime to estimate the Vogel exponent.

Discussion

The results show that the reconfiguration number decreases monotonically with increasing Cauchy number, in agreement with the theoretical behavior described in the literature. The experimental data follow the expected trend, with a rigid regime at low Cauchy numbers and a variable reconfiguration regime at intermediate values. In this regime, a power-law behavior is observed, and the Vogel exponent extracted from the logarithmic slope is $V \simeq -0.28$.

This value is consistent with previously reported results for thin flexible sheets and plates, for which moderate drag reduction due to reconfiguration is observed. In such configurations, the structures are not flexible enough to reach the fully asymptotic regime, leading to Vogel exponents with relatively small negative values. Similar ranges of Vogel exponent have been reported for thin sheets and leaf-like structures in uniform flow [5].

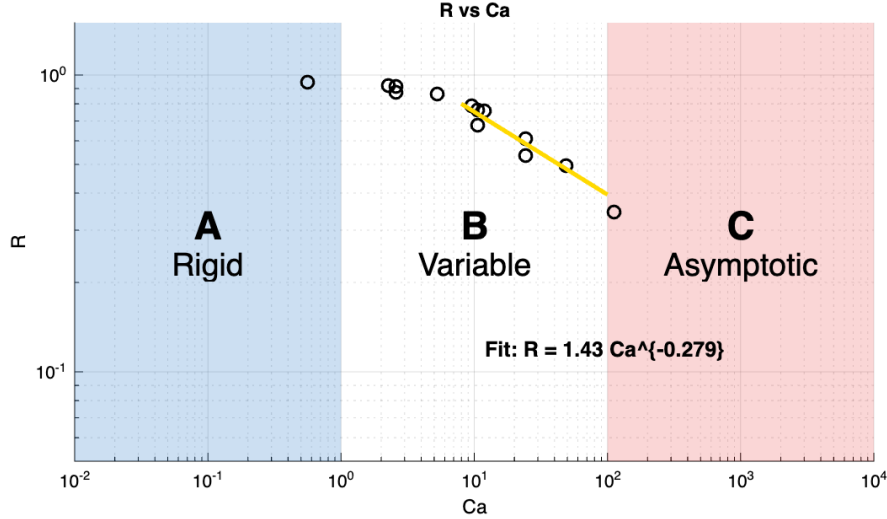


Figure 6: Reconfiguration number as a function of the Cauchy number for all samples.

6.2 Flutter Onset

The onset of flutter was characterized using both qualitative and quantitative approaches. Qualitatively, flutter was visualized by superimposing all frames of a recorded image sequence, allowing the full range of sheet positions to be observed simultaneously. This representation highlights the minimum and maximum bending configurations of the sheet and provides a clear visual indication of the large-amplitude oscillatory motion, as illustrated in Fig. 7.

When the oscillation amplitude becomes sufficiently large, the sheet is considered to be in a flutter regime. In addition to this qualitative criterion, a quantitative measure was obtained by comparing the minimum and maximum values of the effective bent length L_{bent} over an oscillation cycle, as illustrated in Section 8. The difference between these two values provides a measure of the oscillation amplitude, which allows a consistent comparison between different flow conditions and sheet configurations.

Discussion

The results show that the onset of flutter occurs at lower forcing conditions for samples with either smaller thickness or larger length. In particular, for the thinnest sheets ($t = 100 \mu\text{m}$), flutter is observed at relatively low motor frequencies, between 10 and 20 Hz depending on the sheet length. For the shorter sheet ($L = 6 \text{ cm}$), flutter appears around 20 Hz, whereas for the longer sheet ($L = 10 \text{ cm}$), flutter is already observed at 10 Hz. This indicates that increasing the sheet length significantly promotes the development of large-amplitude oscillations.

For thicker sheets ($t = 300 \mu\text{m}$), the onset of flutter occurs at higher motor frequencies due to stronger structural restoring forces. However, a similar trend with respect to sheet length is observed: increasing the length leads to earlier and more pronounced fluttering behavior. These observations are consistent with the expected balance between aerodynamic forcing and elastic stiffness, where thinner and longer sheets are more susceptible to flow-induced instabilities.

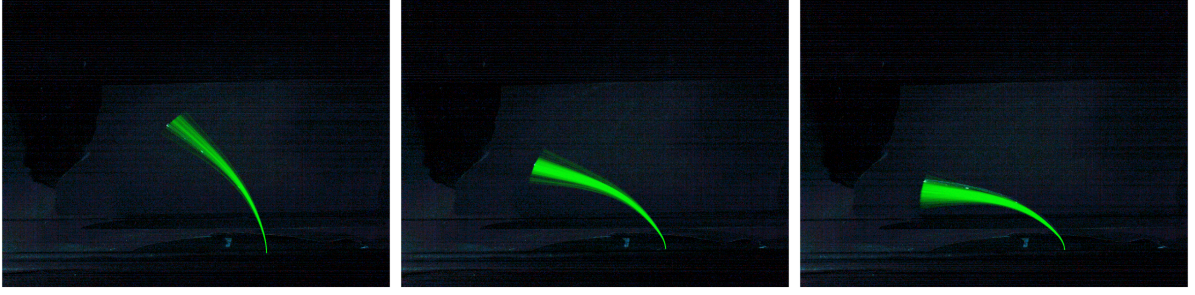


Figure 7: Flutter visualization for the $t = 300 \mu\text{m}$, $L = 10 \text{ cm}$ sheet obtained by frame superposition.

6.3 Frequency Extraction: Fourier Spectrum and Temporal Autocorrelation

The comparison between the frequency extraction methods based on Fourier spectral analysis and temporal autocorrelation is presented in Section 8. The dominant oscillation frequency can be obtained from the temporal autocorrelation by identifying the first peak away from the origin, which corresponds to the characteristic oscillation period.

Discussion

It is observed that the frequency extracted from the Fourier spectrum is generally less accurate than the one obtained from the temporal autocorrelation. The Fourier spectrum identifies the dominant peak in the frequency domain; however, the signals are often noisy, leading to large low-frequency peaks that can obscure the true oscillation frequency. As a result, the dominant peak in the Fourier spectrum does not always correspond to the physical flutter frequency. Nevertheless, in most cases a spectral peak is observed at a frequency close to that obtained from the temporal autocorrelation. The autocorrelation method proves to be more robust, as it relies on the identification of the first peak away from the origin in the time domain, which is less sensitive to low-frequency noise. This makes temporal autocorrelation a more reliable method for frequency extraction in the present experiments. But in configurations where almost no oscillatory motion is observed, such as for the $t = 500 \mu\text{m}$, $L = 6.0 \text{ cm}$ sample at $f_M = 10 \text{ Hz}$, both methods are dominated by measurement noise, leading to large errors in the extracted frequencies that do not correspond to a physical flutter mode (Fig. 14).

6.4 Strouhal Number and Comparison with Natural Frequency

In this work, two Strouhal numbers are computed using the frequencies extracted from Fourier spectral analysis (f_{FFT}) and temporal autocorrelation (f_{AC}). The Strouhal number is directly proportional to the frequency.

To compare the measured flutter frequencies with the intrinsic structural dynamics, the natural frequency of the sheet is estimated using the Euler–Bernoulli cantilever-beam approximation (clamped–free). The fundamental natural frequency is

$$f_n = \frac{\beta_1^2}{2\pi} \sqrt{\frac{EI}{\rho_s AL^4}}, \quad \beta_1 \approx 1.875, \quad (8)$$

where the expression for the natural frequency and the modal constant β_1 are taken from classical Euler–Bernoulli beam theory [1], E is the Young’s modulus, I the second moment of area of the rectangular cross-section ($I = bh^3/12$), we assume the density to be $\rho_s \approx 1300 \text{ kg/m}^3$ and $A = bh$ the cross-sectional area. When the measured flutter frequency is close to f_n , the dynamics are primarily structure-dominated, whereas significant deviations indicate a stronger influence of flow forcing.

Table 3: Natural frequencies of the sheets for each thickness and length configuration.

t (μm)	L (cm)	f_n (Hz)
300	6	17.91
300	10	6.45
500	6	29.84
500	10	10.74

Discussion

The comparison between the measured flutter frequencies and the natural frequencies shows that, at low flow velocities, the oscillation frequency is of the same order of magnitude as the natural frequency, indicating a structure-dominated regime. As the flow velocity increases, the flutter frequency departs from the natural frequency, revealing a transition toward a flow-dominated regime. This transition from a structure-dominated regime, where the oscillation frequency is close to the natural frequency, to a flow-dominated regime at higher flow velocities has been reported for flexible structures interacting with fluid flows [3].

7 Conclusion

In this experiment, the fluid–structure interaction of flexible sheets subjected to an airflow was investigated through analyses of static reconfiguration and dynamic fluttering. The results show that the Cauchy number governs the reconfiguration behavior, while the onset and the frequency of flutter depend on both the structural properties of the sheets and the flow velocity. A clear transition from structure-dominated to flow-dominated oscillatory regimes was observed, and temporal autocorrelation proved to be more robust than Fourier spectral analysis for frequency extraction in noisy or non-stationary conditions.

Possible improvements to the experiment include using finer increments of the wind tunnel motor frequency in order to better capture the transitions between different dynamical regimes. In addition, testing a wider range of sheet lengths would help to further characterize the overall evolution of the reconfiguration and flutter behavior and improve the understanding of scaling effects in flexible structures interacting with fluid flows.

References

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8 Appendix

Sheet thickness $300\ \mu\text{m}$ and length $60\ \text{mm}$

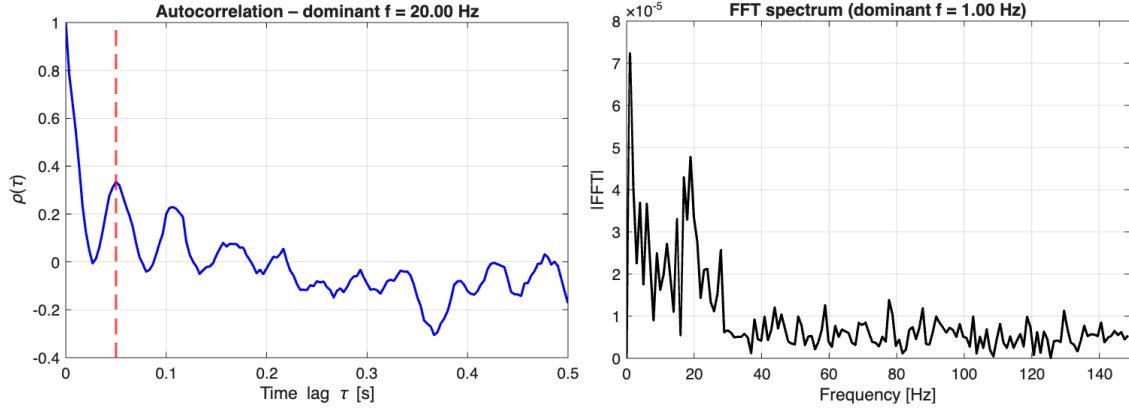


Figure 8: Spectrum and Autocorrelation for the sheet: $300\ \mu\text{m}$, $60\ \text{mm}$ at $10\ \text{Hz}$

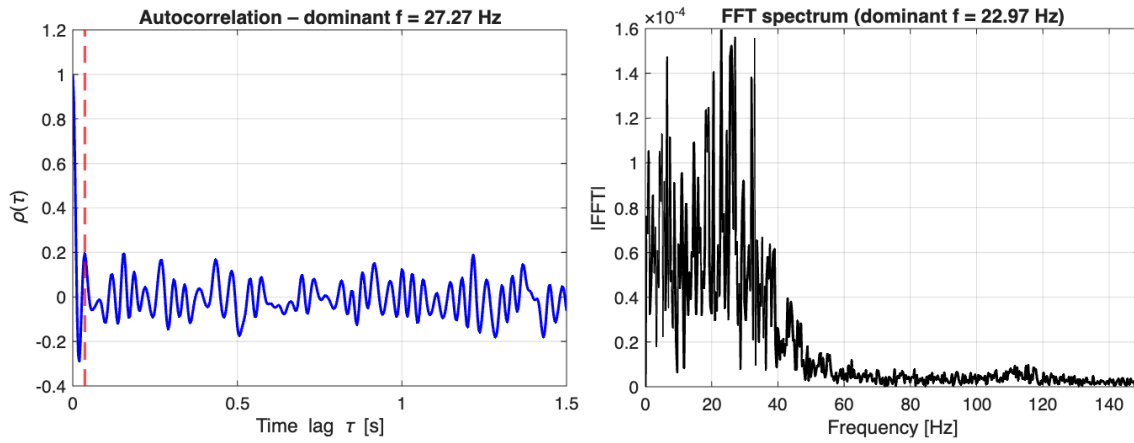


Figure 9: Spectrum and Autocorrelation for the sheet: $300\ \mu\text{m}$, $60\ \text{mm}$ at $20\ \text{Hz}$

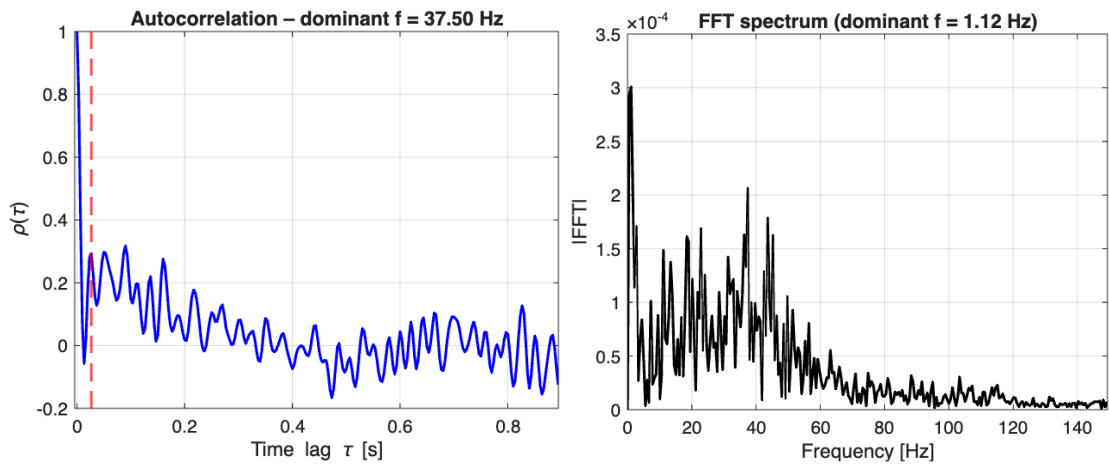


Figure 10: Spectrum and Autocorrelation for the sheet: $300\ \mu\text{m}$, $60\ \text{mm}$ at $30\ \text{Hz}$

Sheet thickness $300\ \mu\text{m}$ and length $100\ \text{mm}$

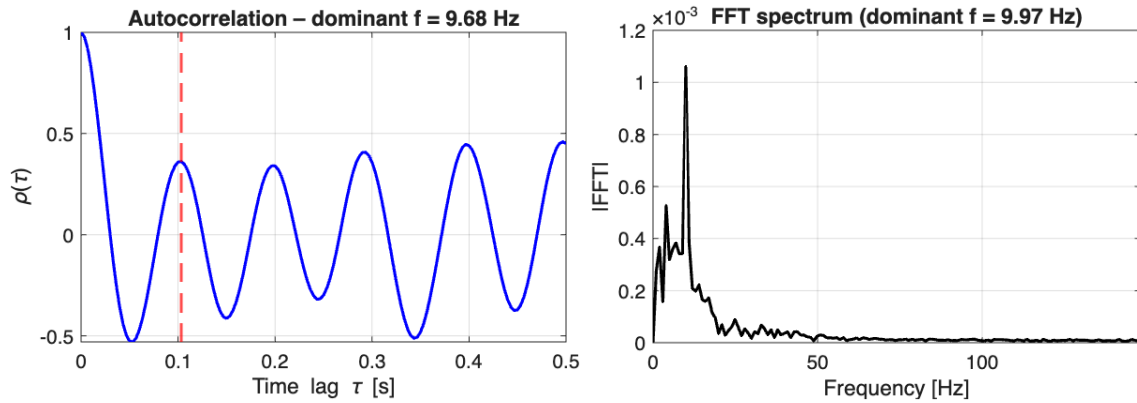


Figure 11: Spectrum and Autocorrelation for the sheet: $300\ \mu\text{m}$, $100\ \text{mm}$ at $10\ \text{Hz}$

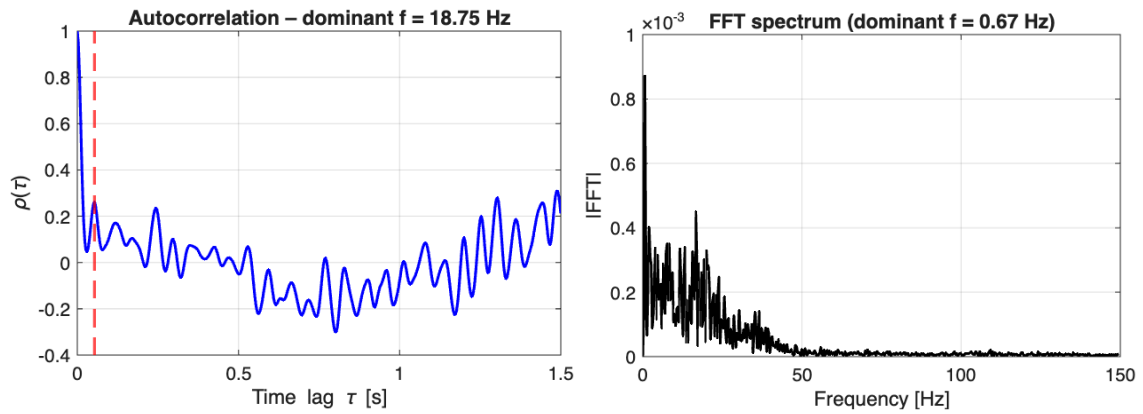


Figure 12: Spectrum and Autocorrelation for the sheet: $300\ \mu\text{m}$, $100\ \text{mm}$ at $20\ \text{Hz}$

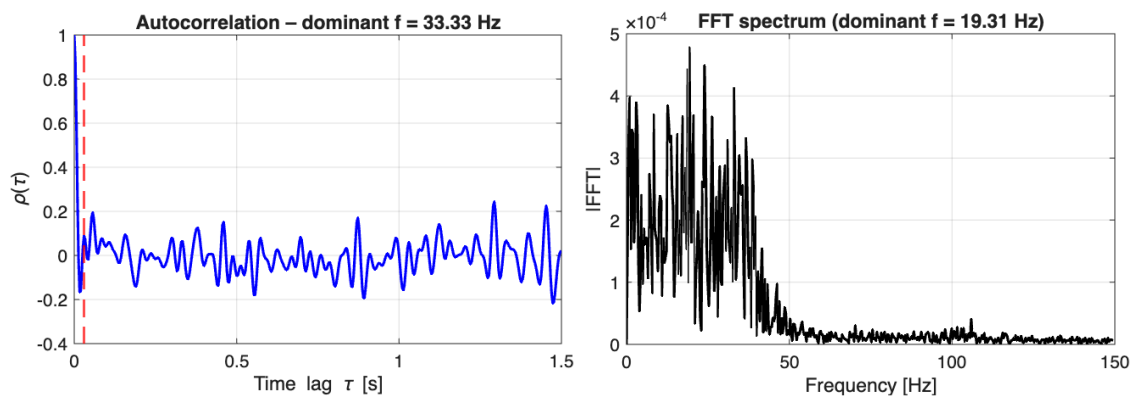


Figure 13: Spectrum and Autocorrelation for the sheet: $300\ \mu\text{m}$, $100\ \text{mm}$ at $30\ \text{Hz}$

Sheet thickness $500\ \mu\text{m}$ and length $60\ \text{mm}$

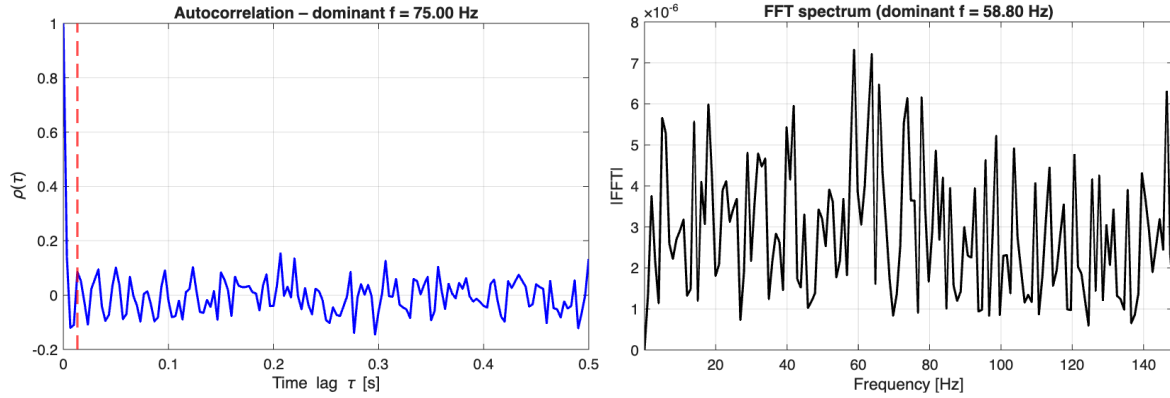


Figure 14: Spectrum and Autocorrelation for the sheet: $500\ \mu\text{m}$, $60\ \text{mm}$ at $10\ \text{Hz}$

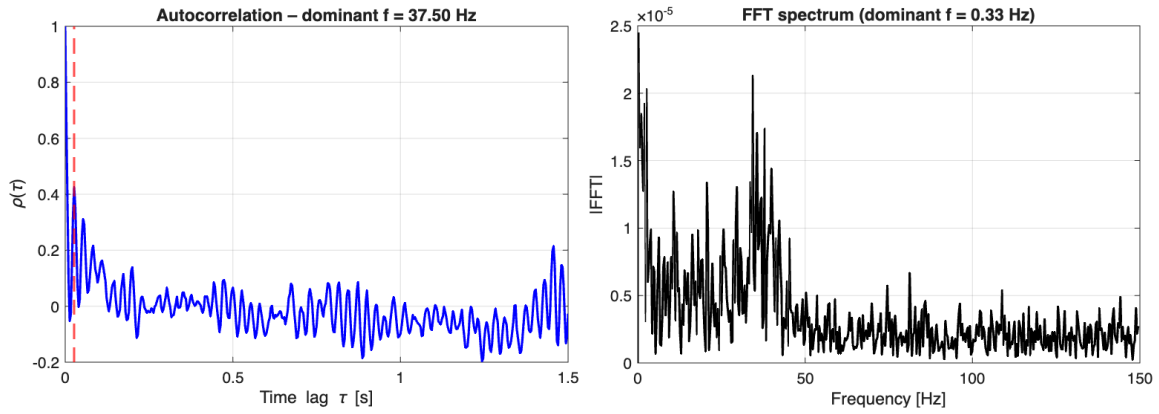


Figure 15: Spectrum and Autocorrelation for the sheet: $500\ \mu\text{m}$, $60\ \text{mm}$ at $20\ \text{Hz}$

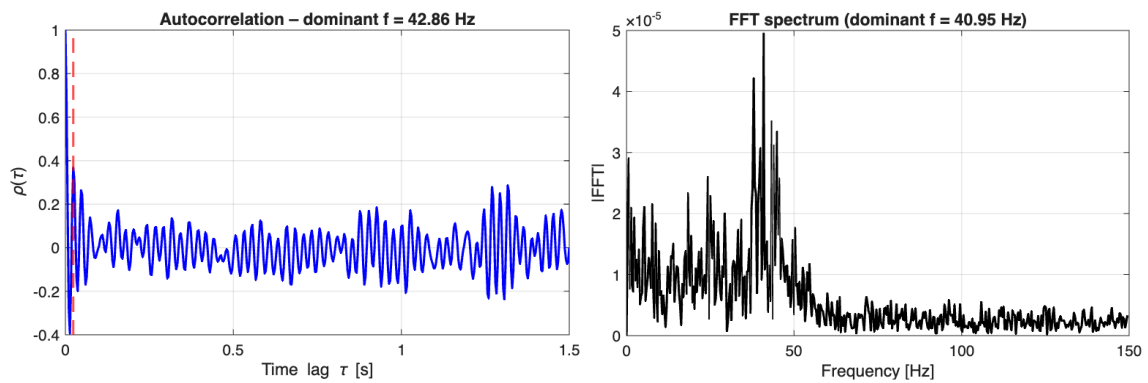


Figure 16: Spectrum and Autocorrelation for the sheet: $500\ \mu\text{m}$, $60\ \text{mm}$ at $30\ \text{Hz}$

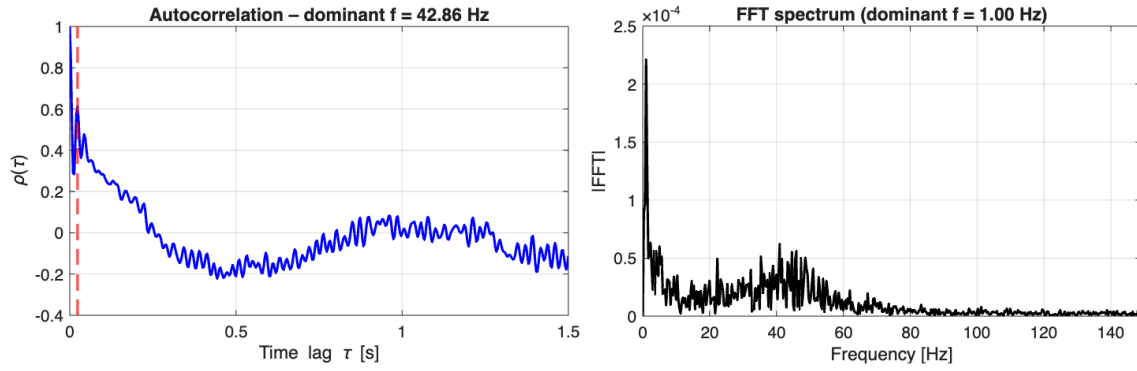


Figure 17: Spectrum and Autocorrelation for the sheet: 500 μm , 60 mm at 40 Hz

Sheet thickness 500 μm and length 100 mm

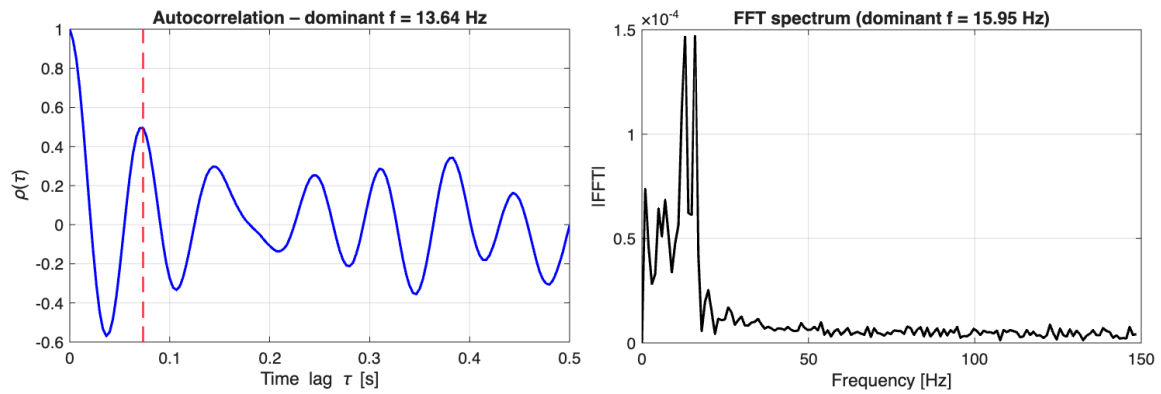


Figure 18: Spectrum and Autocorrelation for the sheet: 500 μm , 100 mm at 10 Hz

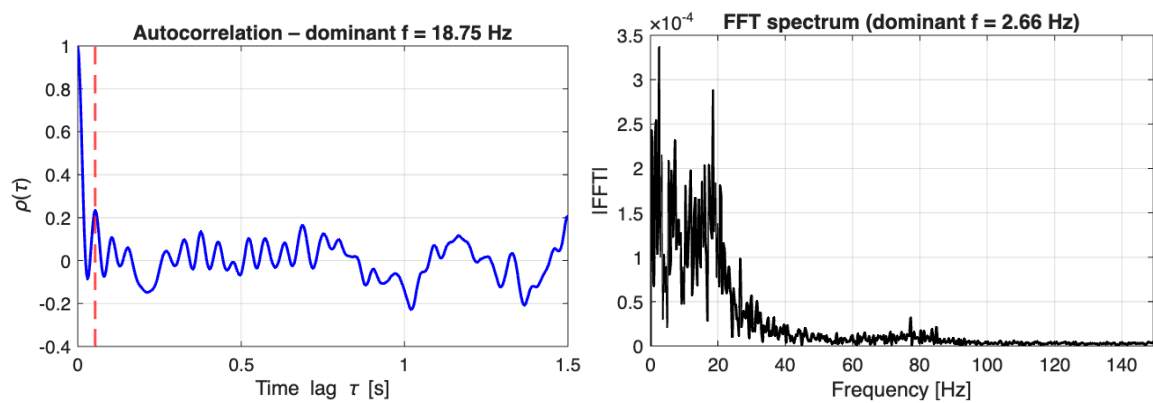


Figure 19: Spectrum and Autocorrelation for the sheet: 500 μm , 100 mm at 20 Hz

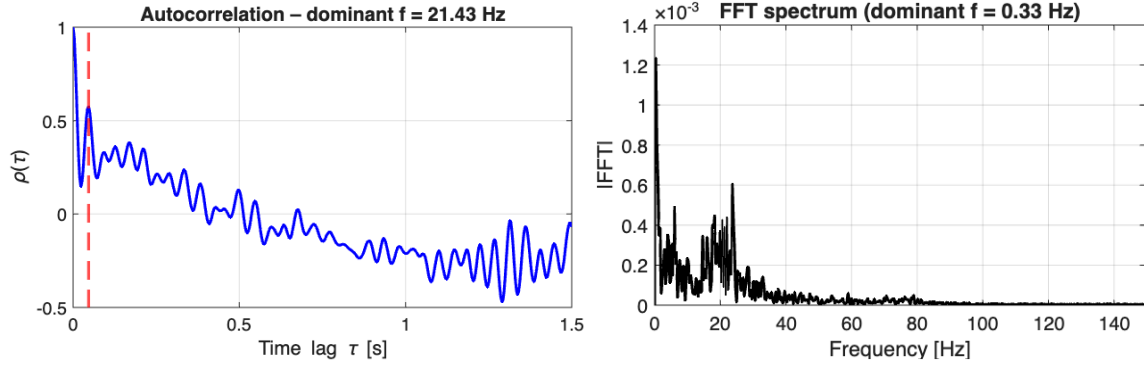


Figure 20: Spectrum and Autocorrelation for the sheet: $500 \mu\text{m}$, 100 mm at 30 Hz

Flutter

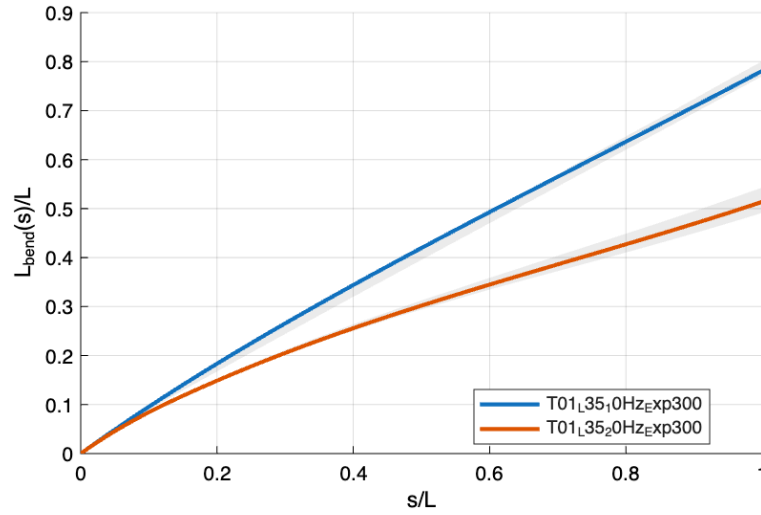


Figure 21: Measured flutter for the $t = 100 \mu\text{m}$, $L = 3.5 \text{ cm}$ sheet

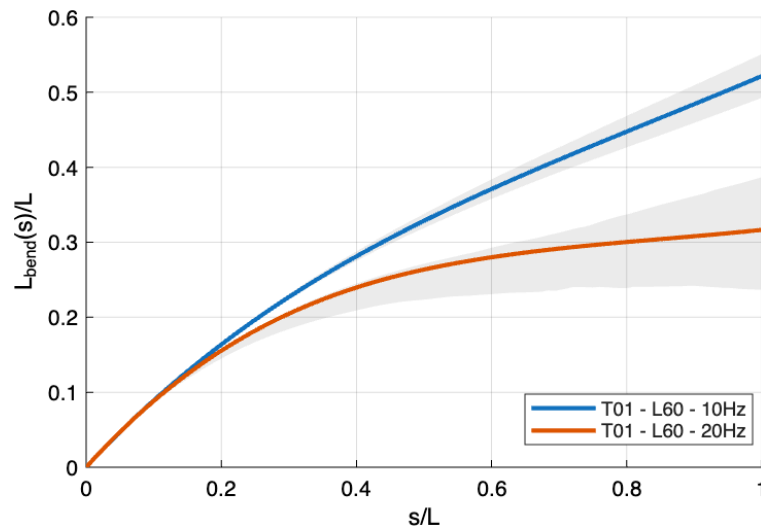
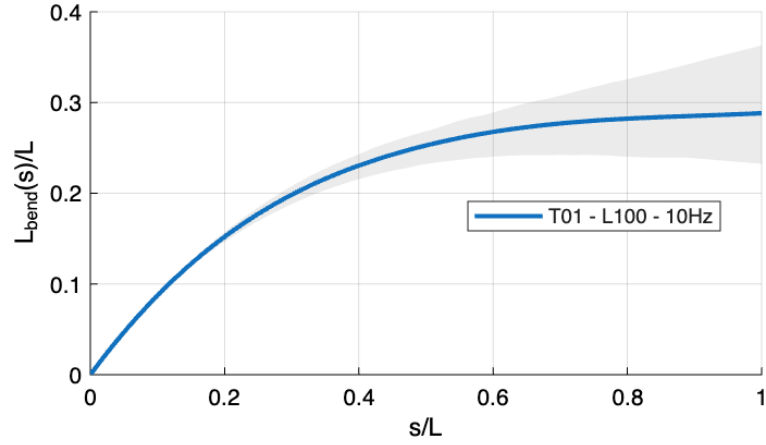
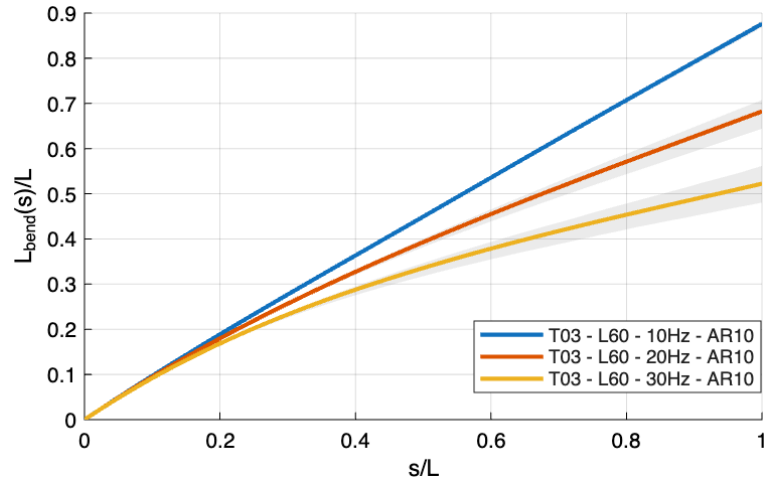
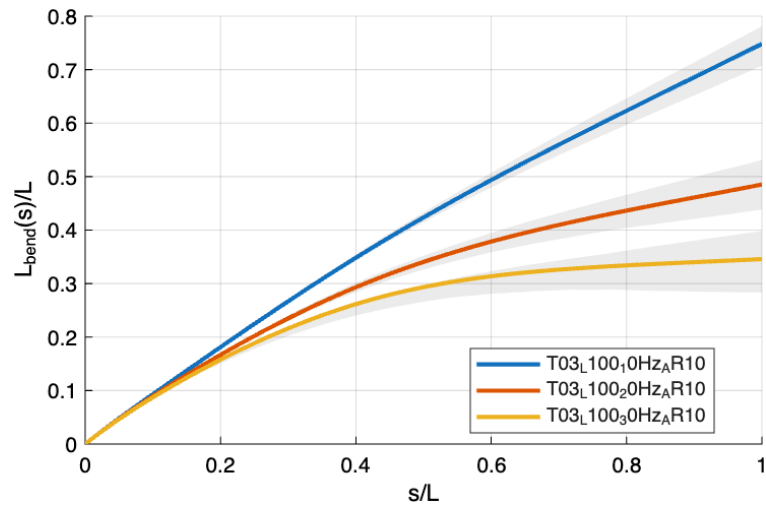


Figure 22: Measured flutter for the $t = 100 \mu\text{m}$, $L = 6 \text{ cm}$ sheet

Figure 23: Measured flutter for the $t = 100 \mu\text{m}$, $L = 10 \text{ cm}$ sheetFigure 24: Measured flutter for the $t = 300 \mu\text{m}$, $L = 6 \text{ cm}$ sheetFigure 25: Measured flutter for the $t = 300 \mu\text{m}$, $L = 10 \text{ cm}$ sheet